

COMPETITIVE PRODUCT ANALYSIS

BALEAF WOMEN'S THERMAL LONG SLEEVE

PURPOSE

This study evaluates the performance characteristics of the Baleaf women's thermal long sleeve through standardized textile testing. In addition to our results presented here, our testing assessed tear strength, colorfastness to crocking, taber abrasion resistance, pilling, wrinkle recovery, and other ASTM exams to determine its effectiveness as a performance garment and adjustments to be considered

PRODUCT DESCRIPTION

The Baleaf women's thermal long sleeve shirt features raglan sleeves, contouring princess seams, an in-seam zipper pocket, thumb loops, and is fleece-lined making it suitable for colder climates. This garment is made to be worn for running or as workout gear.



CONTENT

Shell: 75% polyester, 25% spandex
Pockets: 90% polyester, 10% spandex

In testing, we learned that the garment's performance characteristics can primarily be attributed to its synthetic content.



burn test results

METHODS

Our team used sizes S-XL to test how 4 treatments behaved:

S - Unlaundered, M - Laundered,
L - Softener, XL - Hang Dried

We ran test using samples from each treatment in order to evaluate how laundering affects performance.



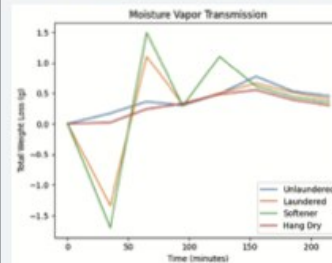
WICKING RATE

Specimen	20 mm	40 mm	Average VW
U Courses	5.36	10.64	3.759
L Courses	1.76	16.9	2.36
S Courses	1.16	11.45	3.49
HD Courses	5.43	6.41	6.25

in seconds

Vertical wicking relates to moving moisture away from the skin, while horizontal wicking spreads it for evaporation. Our tests determined that the garment was effective at quickly absorbing water.

MOISTURE VAPOR TRANSMISSION



MVT measures the permeability of materials to water, crucial for performance garments intended to stay dry.

While the garment was effective at evaporating moisture, the unlaundered and hang dried performed the best.

WATER REPELLENCY

0 - Soaked 5 - No Absorption

The spray rating did not show major differences since all samples were rated 0, indicating poor water resistance and easy wetting. This garment is not suitable to wear in the rain or very wet conditions.

Specimen	Spray Rating
Unlaundered	0
Laundered	0
Softener	0
Hang Dried	0

THERMAL INSULATION

Our testing reveals that this garment is an effective insulator. The laundered sample has the lowest bulk density and might transmit the least heat.

Characteristic	U	L	S	HD
Weight	4.55 g	4.72 g	4.6 g	4.55 g
Thickness	88 mm	88 mm	88 mm	88 mm
Area	12.7 cm ²	12.7 cm ²	12.7 cm ²	12.7 cm ²
Bulk Density	24.56	23.68	24.3	24.56
% Air Content	15.93%	15.35%	15.75%	15.93%

CONCLUSION

The Baleaf women's thermal long sleeve is overall successful at meeting its intended use. The garment is both quick drying to absorb sweat and insulating for optimal performance in cold conditions. Moving forward, the quality assurances team should aim to unify test results across multiple laundering conditions in order to improve garment longevity.